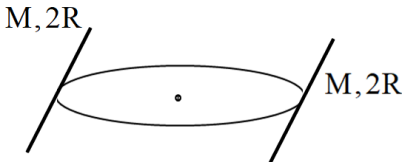
PHYSICS

1) A solid sphere and a hollow sphere of equal mass and radius are placed over a rough horizontal surface after rotating them about centre of mass with same angular velocity ω_0. Once the pure rolling start let V_1 and V_2 be the linear speeds of their centre of mass then ?

(1) $V_1 = V_2$
(2) $V_1 > V_2$
(3) $V_1 < V_2$
(4) Data is insufficient

2) Two rods of mass M and length 2R are connected at the edge of a ring of mass of M and radius R in the plane of ring as shown in the fig. Find the moment of Inertia of the system about centre of the ring and perpendicular to the plane of the ring.



(1) $\frac{11}{3}MR^2$
(2) $\frac{12}{5}MR^2$
(3) $\frac{19}{12}MR^2$
(4) $\frac{7}{3}MR^2$

3) A square plate has side 2m. Its mass per unit area is $3kg/m^2$. Its moment of inertia about a side is :

(1) 4
(2) 12
(3) 16
(4) 15

4) If the radius of a solid sphere is 35cm, calculate the radius of gyration when the axis is along a tangent :

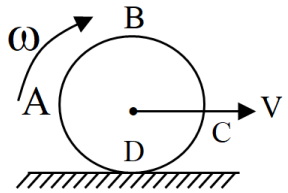
(1) $7\sqrt{35}\,cm$
(2) $7\sqrt{10}\,cm$
(3) 7/5 cm
(4) 2/5 cm

5) The diameter of a flywheal increase by 2.1% and mass is decreased by 0.5% then what will be the

percentage increase in moment of inertia about axis of symmetry ?

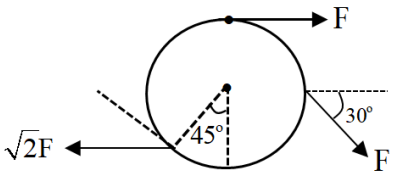
(1) 3.7
(2) 2.1
(3) 2
(4) 1.6

6) A ring rolls without slipping on a horizontal surface. At any instant, its position is shown in the figure. Then which of the following is correct :-



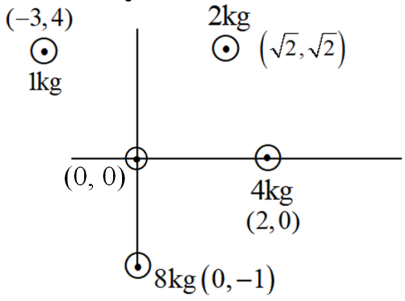
(1) Section AB has less kinetic energy than section CD.
(2) Section BC has more kinetic energy than CD.
(3) Section ABC has the same kinetic energy as section ADC
(4) Acceleration of point D is zero

7) Find the Net torque acting on the ring



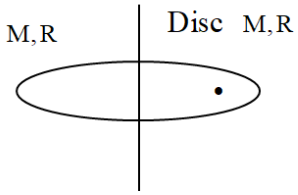
(1) 3FR
(2) $\frac{5F}{2}R$
(3) $\frac{3FR}{\sqrt{2}}$
(4) $\left(2F+\frac{F}{\sqrt{2}}\right)R$

8) Find the moment of inertia of the system about the axis passing through the origin and perpendicular to the plane :



(1) 24
(2) 53
(3) 62
(4) 57

9) A disc of mass M and radius R is revolving with angular velocity ω. A turtle is revolving with the



disc. It start moving along the diameter and reaches to the other end then

(1) Angular velocity of the system first increases and then decreases.

(2) Angular velocity of the system first decreases and then increases.

(3) Angular velocity remains constant.

(4) Angular velocity continuos decreases.

10) **Statement-I :** A ladder is more likely to slip when a person is near the top than when he is near the bottom.
Statement-II : The friction between the ladder and the floor decreases as he climbs up.

(1) Statement-I & II are true and II is the correct explanation of I

(2) Statement-I & II are true but II is not the correct explanation of I

(3) Statement-I is true but II is false

(4) Both statements are false

11) **Assertion (A):** If moment of inertia of a uniform thin circular ring is same about two different axes parallel to each other and lying in the plane of ring, then both the axis can be at same distance from centre of mass of ring.
Reason (R): From parallel axis theorem $I_1 = I_2$ is possible only if $I_{cm} + Md_1^2 = I_{cm} + Md_2^2$

(1) Assertion and Reason both are true and Reason is the correct explanation of Assertion.

(2) Assertion and Reason both are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true and Reason is false

(4) Assertion and Reason both are false

12) A uniform cube of side a and mass M rest on a rough horizontal table. A horizontal force F is applied normal to one of the faces at a point directly above the centre of the face at a height $\frac{2a}{3}$ above the base. The minimum value of F for which the cube begins to flip about the edge is.

(1) $\frac{3Mg}{4}$

(2) $\frac{2Mg}{3}$

(3) $\frac{Mg}{3}$

(4) $\frac{Mg}{4}$

13) A child is standing with folded hands at the centre of a platform which is rotating about its central axis. The kinetic energy of the system is E. The child now stretches his arms so that the moment of inertia of the system becomes four times. Then the kinetic energy of the system now is

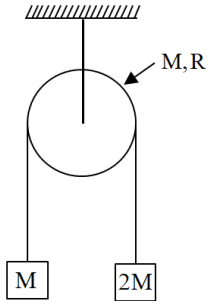
(1) E

(2) E/2

(3) E/4

(4) 4E

14) Two particles of mass 2M & M are tied with two ends of a string, which passes over a rough pulley as shown in the diagram. Assuming pulley as a ring & string does not slip on pulley, find the acceleration of block M



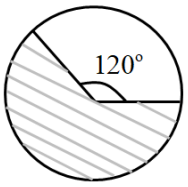
(1) $\frac{g}{3}$

(2) $\frac{g}{4}$

(3) $\frac{g}{2}$

(4) $\frac{2g}{3}$

15) From a circular disc of mass M and radius R, a part of 120° is removed as shown. The moment of inertia of the remaining portion of the disc about an axis passing through the center and perpendicular to the plane of disc.



(1) $\frac{MR^2}{2}$

(2) $\frac{MR^2}{3}$

(3) $\frac{MR^2}{8}$

(4) $\frac{3MR^2}{5}$

16) A wheel initially at rest is rotated with uniform angular acceleration. The wheel rotates through an angle θ_1 in first five seconds and through an additional angle θ_2 in next 20 sec. Then ratio of $\frac{\theta_2}{\theta_1}$ is.

(1) 9

(2) 12

(3) 36

(4) 24

17) An inclined plane makes an angle 60° from horizontal. A solid sphere rolls without slipping on it. The linear acceleration of the sphere will be.

(1) $\frac{2\sqrt{3}}{14}g$

(2) $\frac{5\sqrt{3}}{14}g$

(3) $\frac{5}{14}g$

(4) $\frac{3}{14}g$

18) A particle of mass, m is thrown with speed V_0 at angle 53° with the horizontal. The angular momentum about the point of projection when the particle is at the highest point of its trajectory.

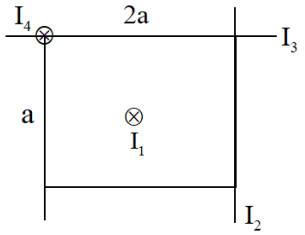
(1) $\frac{\sqrt{3}mV_0^3}{16g}$

(2) $\frac{12mV_0^3}{125g}$

(3) $\frac{17mV_0^3}{40g}$

(4) $\frac{24mV_0^3}{125g}$

19) A rectangular plate of mass M and side a and 2a is given. Different axis of rotations are marked.



Match the column :

	Column-I		Column-II
(a)	I_1	(P)	$\frac{4Ma^2}{3}$
(b)	I_2	(Q)	$\frac{5Ma^2}{12}$
(c)	I_3	(R)	$\frac{5Ma^2}{3}$
(d)	I_4	(S)	$\frac{Ma^2}{3}$

(1) a→P, b→S, c→R, d→Q

(2) a→S, b→Q, c→R, d→Q

(3) a→Q, b→R, c→P, d→S

(4) a→Q, b→P, c→S, d→R

20) Match the following

	Current		r.m.s values
(A)	$x_0 \sin\omega t$	(i)	x_0
(B)	$x_0 \sin\omega t \cos\omega t$	(ii)	$\frac{x_0}{\sqrt{2}}$
(C)	$x_0 \sin\omega t + x_0 \cos\omega t$	(iii)	$\frac{x_0}{(2\sqrt{2})}$

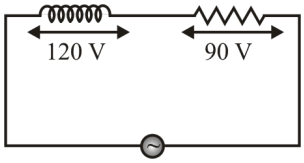
(1) (A → i), (B → ii), (C → iii)

(2) (A → ii), (B → iii), (C → i)

(3) (A → i), (B → iii), (C → ii)

(4) None of these

21) If current in the circuit is 3A, then impedance of the circuit shown in the figure is :-



(1) 100 Ω

(2) 50 Ω

(3) 30 Ω

(4) 40 Ω

22) Voltage and current in an ac circuit are given by V = $5\sin\left(100\pi t - \frac{\pi}{6}\right)$ and I = $4\sin\left(100\pi t + \frac{\pi}{6}\right)$. Then :-

(1) Voltage leads the current by 30°

(2) Current leads the voltage by 30°

(3) Current leads the voltage by 60°

(4) Voltage leads the current by 60°

23) An ac source is rated at 220V, 50 Hz. The time taken for voltage to change from its peak value to zero is

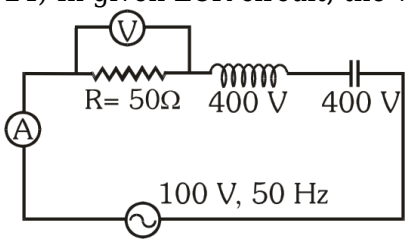
(1) 50 sec

(2) 0.02 sec

(3) 5 sec

(4) 5×10^{-3} sec

24) In given LCR circuit, the voltage across the terminals of a resistance & current will be–



(1) 400V, 2A

(2) 800V, 2A

(3) 100V, 2A

(4) 100V, 4A

25) If V = 100 sin100t volt, and I = 100 sin$(100t + \frac{\pi}{6})$ A. then find the watt less power in watt :-

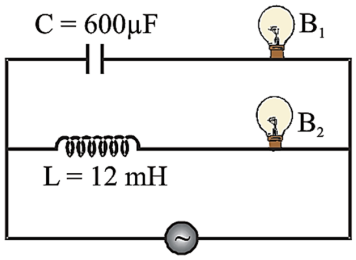
(1) 10^4

(2) 10^3

(3) 10^2

(4) 2.5×10^3

26) In the circuit shown in figure, if both the bulbs B_1 and B_2 identical then :-



(1) Both bulbs have equal brightness.

(2) B_2 will be brighter than B_1.

(3) As the frequency is increased, brightness of B_1 will increase and that of B_2 will decrease.

(4) As the frequency is decreased the brightness of B_1 will increase and that of B_2 will decrease.

27)

What is the frequency of instantaneous power in an AC circuit connected to a capacitor. Given f = 30 Hz, V = 200 V

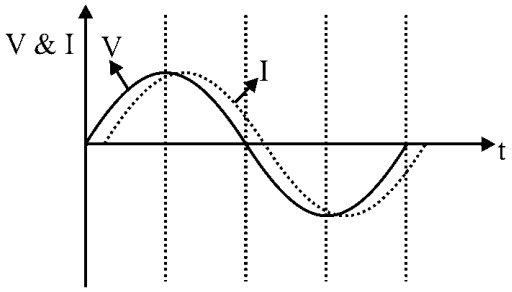
(1) 50 Hz

(2) 100 Hz

(3) zero

(4) 60 Hz

28) Graph shows voltage (V) and current (I) against time for an AC circuit where all elements are in



series. Circuit may contain :-

(1) L, C only

(2) L only

(3) L,R only

(4) R, C only

29) In an ac circuit, the power factor

(1) Is zero when the circuit contains an ideal resistance only

(2) Is unity when the circuit contains an ideal resistance only

(3) Is unity when the circuit contains an ideal inductor only

(4) Is unity when the circuit contains an ideal capacitor only

30) Match the entries given in both the columns, For a series RLC AC circuit -

Column-I		**Column-II**	
(P)	$\omega_{source} > \omega_{resonant}$	(1)	Circuit will be inductor dominant
		(2)	Circuit will be capacitor-dominant
(Q)	$\omega_{source} < \omega_{resonant}$	(3)	Neither inductor, nor capacitor will be dominant in circuit
		(4)	Current will lead voltage
(R)	$\omega_{source} = \omega_{resonant}$	(5)	Current will lag voltage
		(6)	Current will be in phase with voltage

(1) P-2,5 ; Q-1,6 ; R-3,4

(2) P-1,5 ; Q-2,4 ; R-3,6

(3) P-2,6 ; Q-1,4 ; R-3,5

(4) P-1,5 ; Q-3,6 ; R-2,4

31) **Assertion (A):** In a series R-L-C circuit, the voltages across resistor, inductor and capacitor are 8V, 16V and 10V, respectively. The resultant emf in the circuit is 10V.

Reason (R): Resultant emf of the circuit is given by the relation, $E = \sqrt{V_R^2 + (V_L - V_C)^2}$

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

32) The Q factor of a series LCR circuit with L = 2H, C = 32μF and R = 10 Ω is :-

(1) 15
(2) 20
(3) 25
(4) 30

33) The electric field intensity produced by the radiations coming from 100 W bulb at a 3 m distance is E. The electric field intensity produced by the radiations coming from 50 W bulb at the same distance is :

(1) $\frac{E}{2}$
(2) 2E
(3) $\frac{E}{\sqrt{2}}$
(4) $\sqrt{2}E$

34) The magnetic field of a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{j}$ then the associated electric field will be :

(1) $3 \times 10^{-8} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{i}$ V/m
(2) $3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{i}$ V/m
(3) $9 \sin(1.6 \times 10^3 x - 48 \times 10^{10} t)\, \hat{k}$ V/m
(4) $9 \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{k}$ V/m

35) Electromagnetic wave consists of periodically oscillating electric and magnetic vectors :

(1) in mutually perpendicular planes but vibrating with a phase difference of π
(2) in mutually perpendicular planes but vibrating with a phase difference of $\frac{\pi}{2}$
(3) in randomly oriented planes but vibrating in phase
(4) in mutually perpendicular planes but vibrating in phase

36) A plane electromagnetic wave of wave intensity 6 W/m^2 strikes a small mirror of area 40 cm^2, held perpendicular to the approaching wave. The momentum transferred by the wave to the mirror each second will be :-

(1) 6.4×10^{-7} kg–m/s
(2) 4.8×10^{-8} kg–m/s
(3) 3.2×10^{-9} kg–m/s
(4) 1.6×10^{-10} kg–m/s

37) Power of point source is 18 mW. Find the intensity 2.5 m away from the source.

(1) 0.23 mW/m^2

(2) 9 mW/m^2

(3) 2.3 W/m^2

(4) 2.88 mW/m^2

38) An unpolarized light of average intensity I is passed through a polarizer. Find the amplitude of electric field of light coming out.
(Symbols have their usual meaning) :–

(1) $\sqrt{\frac{2I}{c\in_o}}$

(2) $\sqrt{\frac{I}{c\in_o}}$

(3) $\sqrt{\frac{4I}{c\in_o}}$

(4) $\sqrt{\frac{4I}{\sqrt{3}\,c\in_o}}$

39) The conduction current is the same as displacement current in capacitor circuit when source is :-

(1) ac only

(2) dc only

(3) both ac and dc

(4) neither dc nor ac

40) In the following sequences, decreasing order of wavelength of EM radiations is :-

(1) Infra-red waves, Radio waves, X-rays, Visible light

(2) Radio waves, Infra-red waves, Visible light , X-rays

(3) Radio waves, Visible light, Infra-red waves, X-rays

(4) X-rays, Visible light, Infra-red waves, Radio waves

41) For a plane electromagnetic wave direction of electric field (E) and magnetic field (B) is along $-\hat{i}+\hat{j}$ and $-\hat{i}-\hat{j}$ respectively. The direction of propagation of plane electromagnetic wave is :-

(1) +x direction

(2) +z direction

(3) +y direction

(4) –x direction

42) Match **List-I** with **List-II**. (The symbols carry their usual meaning.)

	List-I		**List-II**
A.	$\oint \vec{E}.d\vec{A} = \frac{Q}{\varepsilon_0}$	I.	Ampere-Maxwell's Law

B.	$\oint \vec{B}.d\vec{A} = 0$	II.	Faraday's Law
C.	$\oint \vec{E}.\vec{dl} = \frac{-d(\phi_B)}{dt}$	III.	Gauss Law of electrostatics
D.	$\oint \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d(\phi_E)}{dt}$	IV.	Gauss Law of magnetism

Choose the **correct** answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-III, B-II, C-IV, D-I

(3) A-IV, B-I, C-III, D-II

(4) A-III, B-IV, C-II, D-I

43) A point source emits radiation at power 31.4 watt. Intensity of radiation at a distance 50 cm from the source will be equal to :

(1) $10\frac{W}{m^2}$

(2) $10^3\frac{W}{m^2}$

(3) $250\frac{W}{m^2}$

(4) $400\frac{W}{m^2}$

44) A uniform meter scale is balanced horizontally on a knife edge at distance of 10 cm from its centre of gravity, when a mass of 50 g and 30 g are suspended from the 5 cm and 90 cm mark on scale. The weight of scale is :

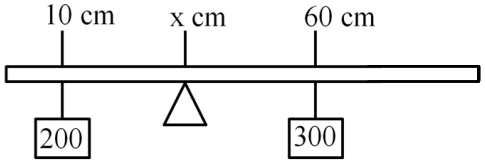
(1) 25 gram-weight

(2) 30 gram-weight

(3) 35 gram-weight

(4) 50 gram-weight

45) A meter scale of negligible mass, balances about a knife edge when mass of 200 g and 300 g are suspended from 10 cm and 60 cm mark respectively. The position of knife edge will be:



(1) 40 cm

(2) 45 cm

(3) 50 cm

(4) 55 cm

CHEMISTRY

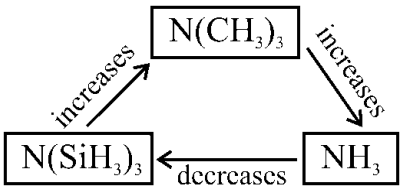
1) Total number of terminal 'H' in B_2H_6 ?

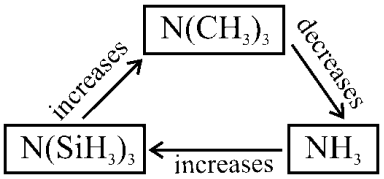
(1) 2
(2) 4
(3) 3
(4) 6

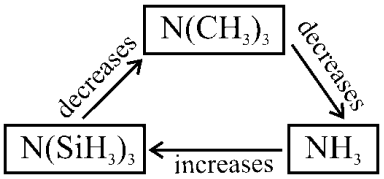
2) Which one of the following gives propyne on treatment with water.

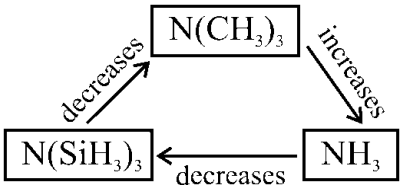
(1) Al_4C_3
(2) CaC_2
(3) Mg_2C_3
(4) Be_2C

3) In which of the following diagram the change in the bond angle at nitrogen is observed correct

(1)


(2)


(3)


(4)


4) Which of the following cannot be hydrolysed ?

(1) NF_3
(2) CCl_4
(3) NCl_3
(4) Both 1 and 2

5) Which of the following halogen form only one oxyacid.

(1) F
(2) Br
(3) I
(4) Cl

6) Which does not contain X–X bond ? (X = Non metal)

(1) $H_2S_2O_5$
(2) $H_2S_2O_6$
(3) $H_2S_2O_7$
(4) $H_4P_2O_6$

7) In which of the following molecule, vacant orbital is not necessary for dimer formation :-

(1) BH_3
(2) $AlCl_3$
(3) CH_3COOH
(4) $BeCl_2$

8) Which of the following molecule have 3c-4e$^-$ bond ?

(1) B_2H_6
(2) $(AlCl_3)_2$
(3) $(BeH_2)_2$
(4) None

9) Which of the following is planar due to back bonding :-

(1) CH_4
(2) $N(CH_3)_3$
(3) $N(SiH_3)_3$
(4) PF_3

10) The correct order of solubility in water for He, Ne, Ar, Kr, Xe is

(1) He > Ne > Ar > Kr > Xe
(2) Ne > Ar > Kr > He > Xe
(3) Xe > Kr > Ar > Ne > He
(4) Ar > Ne > He > Kr > Xe

11) In $Na_2[Fe(CN)_5NO]$, sodium nitroprusside :-

(1) Oxidation state of Fe is +2
(2) It has NO^+ as ligand
(3) Both are correct

(4) None is correct

12) The oxidation no and coordination no. of Pt in $[Pt(C_2H_4)Cl_3]^-$:-

(1) +1, 3
(2) +2, 4
(3) +3, 6
(4) +2, 5

13) In $[Fe(NH_3)_6]Cl_3$ the number of covalent bonds is -

(1) 3
(2) 6
(3) 9
(4) 18

14) Identify the incorrect statements for the behaviour of ethane-1,2-diamine as a ligand :-

(1) It is a neutral ligand
(2) It is a bidentate ligand
(3) It is a chelating ligand
(4) It is a unidentate ligand

15) Which of the following are the limitation of valence bond theory.

(1) It does not distinguish between weak and strong ligands.
(2) It does not give quantitative interpretation of magnetic data.
(3) It does not explain the colour exhibited by coordination compounds
(4) All of these

16) EAN of Cu in $[Cu(NH_3)_4]^{+2}$ is (Cu = 29) :-

(1) 35
(2) 36
(3) 49
(4) 40

17) What fraction of chlorine will be precipitated as AgCl by using $AgNO_3$ from $[CrCl(NH_3)_5]Cl_2$:-

(1) $\frac{1}{3}$
(2) $\frac{2}{3}$
(3) $\frac{1}{4}$
(4) 1

18) The correct IUPAC name for the compound $[Ag(NH_3)_2][Ag(CN)_2]$ is

(1) Diamminesilver(II)dicyanidoargentate(II)
(2) Diamminesilver(I)dicyanidoargentate(II)
(3) Diamminesilver(II)dicyanidoargentate(I)
(4) Diamminesilver(I)dicyanidoargentate(I)

19) Pentaaquachlorochromium(III) chloride monohydrate complex has the formula :-

(1) $[Cr(H_2O)_6]Cl_2$
(2) $[Cr(H_2O)_5Cl]Cl_2 . H_2O$
(3) $[Cr(H_2O)_4Cl_2]Cl . 2H_2O$
(4) None

20) Total no. of isomers in $[Pt(NH_3)_2(NO_2)_2]$ are -

(1) 2
(2) 3
(3) 6
(4) 9

21) Fac and Mer isomerism is associated with which of the following general formula :-

(1) $[M(AA)_2]$
(2) $[M(AA)_3]$
(3) [M ABCD]
(4) $[M A_3B_3]$

22) Which of the following is correct order of cyrstal field splitting.

(1) C–donor > N–donor > O–donor
(2) N–donor > O–donor > C–donor
(3) C–donor > O–donor > N–donor
(4) O–donor > N–donor > C–donor

23) Spin only magnetic moment of $[NiCl_4]^{2-}$ will be

(1) 2.82 B.M.
(2) 0 B.M
(3) 1.23 B.M.
(4) 5.64 B.M.

24) Incorrect statement regarding $[Cr(NH_3)_6]Cl_3$ is :-

(1) d^2sp^3 Hybridisation & octahedral geometry

(2) Paramagnetic complex

(3) Outer orbital complex

(4) Gives white ppt with $AgNO_3$ solution

25) For the complex showing the square planar structure the d-orbital involved in the hybridisation.

(1) $dx^2–y^2$

(2) dz^2

(3) dxy

(4) dxz

26) Geometry of $[PdCl_4]^{-2}$ is:-

(1) octahedral

(2) Tetrahedral

(3) linear

(4) Square planer

27) The magnetic moment of complexes given below are in the order :
(I) $[Ni(CO)_4]$ (II) $[Mn(CN)_6]^{4-}$
(III) $[Cr(NH_3)_6]^{3+}$ (IV) $[CoF_6]^{3-}$

(1) I > II > III > IV

(2) I < II < III < IV

(3) IV > III > I > II

(4) IV > II > I > III

28) Which of the following has sp^3 hybridisation :-

(1) $[Ni(CN)_4]^{-2}$

(2) $[PtCl_4]^{-2}$

(3) $[AgF_4]^{-}$

(4) $[CuCl_4]^{-2}$

29) Which of the following complex represent maximum stability ?

(1) $[Co(CN)_6]^{3-}$

(2) $[Co(H_2O)_6]^{3+}$

(3) $[Co(NH_3)_6]^{+3}$

(4) $[Co(F)_6]^{3-}$

30) Which of the following transition element does not show variable oxidation state?

(1) Sc

(2) Fe

(3) Mn
(4) Co

31) Which type of carbide is not formed by lanthanoids.

(1) LnC_2
(2) Ln_2C_3
(3) LnC_3
(4) Ln_3C

32) Which order is correct for oxidising power ?

(1) $Cr_2O_7^{2-} < VO_2^+$
(2) $VO_2^+ > MnO_4^-$
(3) $VO_2^+ < Cr_2O_7^{2-}$
(4) $Cr_2O_7^{2-} > MnO_4^-$

33) Correct order of stability is :-

(1) $Mn^{+6} > Mn^{+7}$
(2) $Mn^{+2} > Mn^{+3}$
(3) $Cu^{+2}_{(aq)} < Cu^{+}_{(aq)}$
(4) $Cr^{+2} > Cr^{+3}$

34) Which of the following can not be oxidised by acidified $K_2Cr_2O_7$

(1) Br^-
(2) I^-
(3) NO_2^-
(4) Cl^-

35) Which of the following element show +7 oxidation state?

(1) Cl
(2) Mn
(3) Np
(4) All of these

36) When MnO_2 is fused with KOH, a coloured compound is the product & the colour is :-

(1) K_2MnO_4, green
(2) $KMnO_4$, purple
(3) Mn_2O_3, brown
(4) Mn_2O_4, black

37) Highest stable Mn fluoride is MnF_4 where as highest Mn oxide is Mn_2O_7 due to–

(1) O atom is smaller the F

(2) Oxygen has ability to form multiple bonds

(3) Mn^{7+} does not exist

(4) F can not-stabilise higher oxidation states

38) Consider the following transformation

$2CuX_2 \xrightarrow[\text{temp.}]{\text{Room}} 2CuX + X_2 \uparrow$ Reducing agent

Then X^- can be :

(1) F^-, I^-

(2) Cl^-, Br^-

(3) CN^-, I^-

(4) Cl^-, F^-

39) When acidified $K_2Cr_2O_7$ react with H_2O_2 a blue colour solution is obtained. This blue colour is due to the formation of :-

(1) Cr_2O_3

(2) CrO_3

(3) $Cr_2(SO_4)_3$

(4) CrO_5

40) Which of the following do not form amalgam with Hg?

(1) Fe

(2) Co

(3) Ni

(4) All

41) Which of the following ion does not exist ?

(1) $[Cu(CN)_4]^{2-}$

(2) VO_4^{3-}

(3) WO_4^{2-}

(4) CrO_4^{2-}

42) A salt gives violet vapours when treated with concentrated H_2SO_4, it contains

(1) Cl^-

(2) I^-

(3) Br^-

(4) NO_3^-

43) When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with concentrated H_2SO_4, orange red vapours are obtained.These are of the compound :-

(1) chromous chloride

(2) chromyl chloride

(3) chromic chloride

(4) chromic sulphate

44) A mixture of salt give rotten egg smell when treated with dilute H_2SO_4 and which turns lead acetate paper black, then salt contain_____.

(1) SO_3^{-2}

(2) S^{-2}

(3) CH_3COO^-

(4) NO_3^-

45)

	List-I (Basic Radical)		**List-II (Group)**
(P)	Pb^{+2}	(1)	VI group
(Q)	Fe^{+3}	(2)	IV group
(R)	Ni^{+2}	(3)	I group
(S)	Mg^{+2}	(4)	III group

Correct matching is :

	P	**Q**	**R**	**S**
(1)	1	3	2	4
(2)	3	4	2	1
(3)	4	3	2	1
(4)	3	4	1	2

(1) 1

(2) 2

(3) 3

(4) 4

BIOLOGY

1) When was the National Committee for Environmental Planning and Coordination established in India ?

(1) 1972

(2) 1984
(3) 1937
(4) 2002

2) Ecology is basically concerned with how many levels of biological organisation ?

(1) 3
(2) 4
(3) 5
(4) 6

3) Which of the following is not an example of a population ?

(1) Lotus plants in a pond.
(2) All the cormorants in a wetland.
(3) Teakwood trees in a forest tract.
(4) All the animals in a forest.

4) If in a pond there were 20 lotus plants last year and through reproduction 4 new plants are added, taking the current population to 24. Calculate the birth rate:

(1) 0.4 offspring per lotus per year
(2) 0.1 offspring per lotus per year
(3) 0.2 offspring per lotus per year
(4) 0.3 offspring per lotus per year

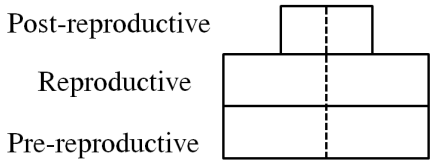
5) **Statement-I :** The size of the population tells us a lot about its status in the habitat.
Statement-II : The size of a population for any species is a static parameter.

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Statement-I and II both are correct.
(4) Statement-I and II both are incorrect.

6) If the age distribution is plotted for the population, the resulting structure is called :

(1) Ecological pyramid
(2) Age pyramid
(3) Sigmoid curve
(4) Exponential curve

7) Given age pyramid represents :



(1) Expanding population

(2) Declining population

(3) Stable population

(4) More pre-reproductive individuals than reproductive individuals

8) Under normal conditions ..... and ..... are the most important factors influencing population density.
Fill in the blanks.

(1) Births, Deaths

(2) Births, Immigration

(3) Deaths, Emigration

(4) Immigration, Emigration

9) Which of the following statement is correct ?

(1) If a new habitat is just being colonised, birth rate may contribute more significantly to population growth than immigration.

(2) Resource availability is obviously essential for the unimpeded growth of a population.

(3) A population growing in a habitat with limited resources show initially acceleration phase.

(4) The geometric growth model is considered a more realistic one.

10)

Find the incorrect match.

(1) Parasitism - Cuscuta on hedge plants

(2) Mutualism-Mycorrhizae

(3) Amensalism- Lichens

(4) Commensalism-Sea anemone and clown fish

11) The most spectacular and evolutionarily fascinating examples of mutualism are found in :-

(1) Animal-animal relationship

(2) Plant-plant relationship

(3) Plant-animal relationship

(4) Plant-fungi relationship

12) **Statement-I :** Mycorrhizae represent an intimate mutualistic relationship between a fungus and photosynthesising algae or cyanobacteria.
Statement-II : Lichens are associations between algae and the roots of higher plants.

(1) Statement-I is correct but Statement-II is incorrect.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Statement-I and II both are correct.

(4) Statement-I and II both are incorrect.

13) Which of the following plant produce thorns as a defence mechanism against predators?

(1) Cuscuta

(2) Acacia

(3) Opium

(4) Calotropis

14) Which of the following statement is incorrect about a predator ?

(1) Predators over-exploits its prey.

(2) Predators keep prey populations under control.

(3) Predators acting as conduits for energy transfer.

(4) Predators help in maintaining species diversity.

15) If a population growing exponentially, double in size after 3 years, than calculate the intrinsic rate of increase.
($\log_e 2 = 0.69314$ & $\log_e 1 = 0$)

(1) 1

(2) 0.3

(3) 0.23

(4) 0.1

16) **Assertion :** Predators are prudent in nature.
Reason : If a predator is too efficient and over-exploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lake of food.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect.

(4) Assertion is incorrect but reason is correct.

17) If two species compete for the same resource, they could avoid competition by choosing different times for feeding or different foraging patterns.
This phenomenon is called :

(1) Competitive exclusion

(2) Competitive release

(3) Interference competition

(4) Resources partitioning

18) A process in which the fitness of one species is significantly lower in the presence of another species is best defined as :

(1) Competition

(2) Mutualism

(3) Commensalism

(4) Protocooperation

19)

In the logistic growth equation given below, the carrying capacity is represented by :

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$$

(1) $\frac{K-N}{K}$

(2) K

(3) N

(4) r

20) Find out the correct match from the following table :

	Column-I	**Column-II**	**Column-III**
(i)	Parasitism	(+, –)	Lichens
(ii)	Competition	(–, –)	Cuscuta on hedge plants
(iii)	Mutualism	(+, +)	Mycorrhizae
(iv)	Commensalism	(+, 0)	Orchid on mango plant

(1) (i) only

(2) (ii) only

(3) (ii) and (iii)

(4) (iii) and (iv)

21)

If a population have higher number of reproductive individuals than the pre-reproductive individuals, then we can predict that population is :

(1) Stable population

(2) Expanding population

(3) Zero growth population

(4) Declining population

22) In general, which of the following appear to be more adversely affected by competition ?

(1) Plants and herbivores

(2) Only plants

(3) Herbivores and carnivores

(4) Only carnivores

23) (i) Epiphyte orchid on mango
(ii) Fig tree and wasp
(iii) Mycorrhizae
(iv) Sea anemone and clown fish
(v) Barnacle on whale
How many of above showing commensalism ?

(1) Two

(2) Three

(3) Four

(4) Five

24) Which of the following act as "Conduits" for energy transfer across trophic levels ?

(1) Decomposer

(2) Predator

(3) Parasite

(4) Producer

25) Which of the following organism is highly distasteful to its predator because of a special chemical present in its body :

(1) Monarch butterfly

(2) Pisaster starfish

(3) Prickly pear cactus

(4) Calotropis

26) In an area, having 200 parthenium plants and a single huge banyan tree, population density is measured in terms of :

(1) Number

(2) Biomass

(3) Percent cover

(4) 2 & 3 both

27) Most important factors influencing population density under normal conditions :

(1) Birth rate, Emigration

(2) Death rate, Immigration

(3) Immigration, Emigration

(4) Birth rate, Death rate

28) Which of the following plant produce highly poisonous cardiac glycosides ?

(1) Cuscuta
(2) Orchid
(3) Parthenium
(4) Calotropis

29) Match the following column-I with column-II.

Column-I		Column-II	
(a)	Camouflaged	(i)	Acacia
(b)	Cardiac glycosides	(ii)	Calotropis
(c)	Thorns	(iii)	Monarch butterfly
(d)	Highly distasteful	(iv)	Frogs

(1) a-(iii), b-(i), c-(ii), d-(iv)
(2) a-(iii), b-(ii), c-(i), d-(iv)
(3) a-(iv), b-(ii), c-(i), d-(iii)
(4) a-(iv), b-(i), c-(ii), d-(iii)

30) Which of the following is an important predator in the rocky intertidal communities of the american pacific coast ?

(1) Pisaster starfish
(2) Barnacle balanus
(3) Abingdon tortoise
(4) Prickly pear cactus

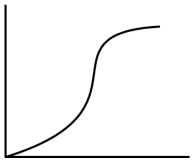
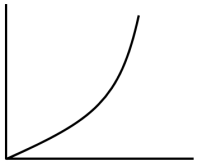
31) In a population showing sigmoid growth form, the population has reached its carrying capacity. What will be the value of $\frac{dN}{dt}$ in this population ?

(1) Equal to K
(2) 1
(3) Zero
(4) Equal to $\left(\frac{K-N}{K}\right)$

32) More recent studies point out that species facing competition might evolve mechanisms that promote co-existence rather than exclusion. One such mechanism is :

(1) Competitive exclusion
(2) Competitive release
(3) Interference competition
(4) Resource partioning

33) Match the following column-I with column-II.

Column-I		Column-II	
(A)	Exponential growth curve	(i)	$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
(B)	Sigmoid growth curve	(ii)	$\frac{dN}{dt} = rN$
(C)	Geometric growth equation	(iii)	
(D)	Logistic growth equation	(iv)	

(1) A-(iv), B-(iii), C-(ii), D-(i)

(2) A-(iv), B-(iii), C-(i), D-(ii)

(3) A-(iii), B-(iv), C-(ii), D-(i)

(4) A-(iii), B-(iv), C-(i), D-(ii)

34) The integral form of exponential growth equation is :

(1) $N_t = N_0 + e^{rt}$

(2) $N_t = N_0 e^{rt}$

(3) $\frac{dN}{dt} = rN$

(4) $N_{t+1} = N_t + [(B + I) - (D + E)]$

35) In how many population interactions only one species get benefitted ?
Predation, Mutualism, Commensalism, Parasitism, Competition, Amensalism

(1) One

(2) Two

(3) Three

(4) Four

36)

Which of the following contributes to decrease in population density ?
(a) Birth rate (b) Death rate (c) Immigration (d) Emigration

(1) a, b

(2) b, c

(3) c, d

(4) b, d

37) What benefits does fig tree offers to wasp in return for its pollination ?

(1) Nectar
(2) Juicy fruits
(3) Pollens
(4) Developing seeds to larvae of wasp

38) In a population birth rate is 0.20 and death rate is 0.08 during a unit time period. What is the value of r (intrinsic rate of natural increase) for given population ?

(1) 0.28
(2) 0.07
(3) 0.12
(4) 0.05

39) What is the reproductive strategy adopted by bamboo plant species to maximise its fitness ?

(1) Breed only once in a life time.
(2) Breed multiple times in a life time.
(3) Produce small number of large sized offsprings.
(4) 2 & 3 both

40) **Assertion (A) :** The problem of predation is particularly severe for the plants.
Reason (R) : Plant cannot run away from their predator unlike animals.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

41) Read the following statements :
(i) Interspecific competition is more severe than intraspecific competition
(ii) Resource need not be limiting for competition to occur as in case of interference competition
(iii) Ecologically, herbivores are predators
(iv) Predators maintain species diversity
Which of the following statements is/are correct ?

(1) (iii) and (iv)
(2) (i) (ii), (iii) and (iv)
(3) (ii), (iii) and (iv)
(4) Only (iv)

42) **Assertion (A) :** Population ecology is an important area of ecology.
Reason (R) : Population ecology links ecology to population genetics and evolution.
In the light of the above statements, choose the correct answer from the option given below.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

43) The following figure shows :



(1) Orchid flower is pollinated by wasp

(2) Orchid flower is pollinated by bee

(3) Fig flower is pollinated by wasp

(4) Fig flower is pollinated by bee

44) To complete its life-cycle human liver fluke depends on :

(1) One intermediate host-snail

(2) One intermediate host-fish

(3) Two intermediate host-snail and fish

(4) Two intermediate host-snail and sheep

45) Many marine fishes are infested with ectoparasitic .....
Fill in the blanks.

(1) Copepods

(2) Ticks

(3) Leech

(4) Barnacles

46) The 'good humor' hypothesis of health was disproved by ?

(1) M.S. Swaminathan

(2) William Harvey

(3) Louis Posteur

(4) Von Behring

47) Health is defined as a state of complete physical, mental and __X__ . What is X ?

(1) Personal hygiene

(2) Balanced diet

(3) Regular exercise

(4) Social well being

48) Which statement is not correct for disease ?

(1) When function of one or more organs of the body is adversely affected and appear by various sign and symptoms.

(2) Disease can be broadly grouped into infectious and contagious.

(3) Disease which are easily transmitted from one person to other are called infectious disease.

(4) Among non-infectious disease cancer is the major cause of death.

49) Among infectious disease which disease is most fatal ?

(1) Cancer

(2) AIDS

(3) Tuberculosis

(4) Hepatitis A

50) Typhoid fever could be confirmed by ?

(1) Widal test

(2) Mantoux test

(3) Dick test

(4) M.P. test

51) Bacteria like streptococcus pneumoniae and haemophilus influenzae in human are responsible for the disease :-

(1) Typhoid

(2) Malaria

(3) Common cold

(4) Pneumonia

52) Which statement is not incorrect for Rhino viruses infection ?

(1) It is usually last in 3–7 days.

(2) It is characterised by nasal congestion and discharge.

(3) They infect the nose respiratory passage and the lungs.

(4) It causes the one of most infectious human ailments.

53) The most serious one and fatal malaria is caused by ?

(1) Plasmodium vivax

(2) Plasmodium malaria

(3) Plasmodium falciparum

(4) Plasmodium ovale

54) In malarial parasite infection the ruptured RBC release a toxic substance that is what ?

(1) Sporozoite
(2) Haemozoine
(3) Trophozoite
(4) Gametocytes

55) Entamoeba histolytica is a protozoan parasite in the _____ of human which causes amoebiasis ?
Fill the blanks ?

(1) Small intestine
(2) Liver
(3) Spleen
(4) Large intestine

56) Which statement is not correct for Amoebiasis ?

(1) Symptoms of this disease include constipation abdominal pain and cramps.
(2) Symptoms of this disease include stools with excess mucous and blood clots.
(3) Mosquito act as vector to transmit the parasite.
(4) Pathogen of this disease is *Entamoeba histolytica.*

57) Match the column-A with column-B correctly ?

	Column-A		**Column-B**
(i)	*Trichophyton*	(a)	Malignant malaria
(ii)	*W. malayi*	(b)	Ringworm
(iii)	*Entamoeba histolytica*	(c)	Filariasis
(iv)	*P. falciparum*	(d)	Amoebiasis

Option :-
(1) i-b, ii-a, iii-d, iv-c
(2) i-a, ii-b, iii-d, iv-c
(3) i-b, ii-c, iii-d, iv-a
(4) i-b, ii-c, iii-a, iv-d

58) The use of vaccines and immunisation programmes have enabled us to __A__ a deadly disease like __B__. Fill in the blanks from option ?

(1) (A)-Partially eradicate (B)-Pollio
(2) (A)-Completely eradicate (B)-Small pox
(3) (A)-Partially eradicate (B)-Small pox
(4) (A)-Completely eradicate (B)-Pneumonia

59) In the given defence which defence is specific type of defence ?

(1) Physical barriers
(2) Physiological barriers
(3) Cell mediated immunity
(4) Cellular barriers

60) **Statement-I :-** Skin on our body is the main barrier which prevents entry of the micro-organisms.
Statement-II :- Acid in the stomach, saliva in the mouth, tears from the eyes all prevents microbial growth.
Find the correct statement :-

(1) Both statement-I and II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statements-I and II are incorrect.

61) Which of the following immune cells performe phagocytosis and destroy microbes.

(1) N.K. cells and T-cells
(2) Basophils and Neutrophils
(3) Neutrophils and Monocytes
(4) Monocytes and T-cells

62) **Statement-I :-** The antibodies are found in the blood that's why it is also called humoral immune response.
Statement-II :- Humoral immune response is one of the type of innate immune response.
Choose the correct option from below options :-

(1) Both statements are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statements are incorrect.

63) When a host is exposed to antigens in the form of vaccine, antibodies are produced in the host body this type of immunity is called :-

(1) Passive immunity
(2) Natural active immunity
(3) Artificial active immunity
(4) Artificial passive immunity

64) When ready made antibodies are directly given to protect the body against foreign agents, it is known as :-

(1) Active immunity

(2) Natural passive immunity
(3) Artificial passive immunity
(4) Artificial active immunity

65) The exaggerated response of the immune system to certain antigens present in the environment is called :-

(1) Active immunity
(2) Allergy
(3) Passive immunity
(4) Vaccine

66) Which type of antibodies is/are produce against the allergens ?

(1) IgG
(2) IgD
(3) IgE
(4) IgM

67) Allergy is due to the release of chemical like __A__ and __B__ from mast cells. A and B are respectively ?

(1) Heparine and histamine
(2) Histamine and serotonin
(3) Heparine and serotonin
(4) Monokines and histamines

68) Transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents, these agents are called ?

(1) Stem cells
(2) Malignant
(3) Mother cells
(4) Carcinogens

69) The lymphoid organs where immature lymphocytes differentiates into antigen sensitive lyphocyte are ?

(1) Lymphnodes
(2) Primary lymphoid organ
(3) Spleen
(4) Secondary lymphoid organ

70) **Statement-(I) :-** Both bone marrow and thymus provide micro environments for the development and maturation of T. lymphocytes.
Statement-(II) :- The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen.

Choose the correct statements :-

(1) Both are correct statements.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both are incorrect.

71) Which lymphoid organ constitutes about 50% of total lymphoid tissue in human body ?

(1) Spleen
(2) MALT (mucosa associated lymphoid tissue)
(3) Thymus
(4) Bone marrow

72) Transmission of HIV infection generally not occurs by :-

(1) Sexual contact with normal person
(2) Transfusion of contaminated blood and blood products.
(3) Sharing infected needles as in the case of intravenous drug abusers
(4) From infected mother to her child through placenta.

73) Which cells progressively decreases in number in the body of AIDS infected person.

(1) B. cells
(2) Neutrophiles
(3) T. helper
(4) Monocytes

74) In the given symptoms which symptoms is not associated with AIDS ?

(1) Weight loss
(2) Diarrhoea
(3) Bouts of fever
(4) Swelling in lower lymphatics

75) The widely used diagnostic test for AIDS is ?

(1) PCR
(2) ELISA
(3) Widal
(4) Mantoux test

76) How many of the listed measures given below help to prevent AIDS ?
(I) Use of disposable syringes
(II) Advocating safe sex
(III) Isolation of AIDS infected person

(IV) Use of condoms during intercourse

(1) I, II and III only
(2) I, II and IV only
(3) II, III and IV only
(4) I, III and IV only

77) Malaria a common disease world wide is caused by a :-

(1) Protozoa
(2) Virus
(3) Bacteria
(4) Helminthes

78) Which of the following immunoglobulins does constitute the largest percentage in human body secretion ?

(1) IgG
(2) IgA
(3) IgD
(4) IgM

79) Heroin is synthesised by ?

(1) Glycosylation of morphine
(2) Acetylation of morphine
(3) Nitration of morphine
(4) Methylation of morphine

80) Match the column-I with column-II correctly :-

	Column-I		**Column-II**
(A)	AIDS	(i)	Hallucinogen
(B)	Smoking	(ii)	Liver cirrhosis
(C)	Alcohol	(iii)	Urinary bladder cancer
(D)	Excessive dose of cocain	(iv)	HIV

(1) A-iii, B-iv, C-i, D-ii
(2) A-iv, B-iii, C-i, D-ii
(3) A-iv, B-iii, C-ii, D-i
(4) A-iv, B-ii, C-iii, D-i

81) MALT constitutes about __X__ percentage of the total lymphoid tissue in human body. What is the percentage of X ?

(1) 50%
(2) 40%
(3) 70%
(4) 30%

82) Liver cirrhosis is caused by the chronic intake of :-

(1) Opium
(2) Alcohol
(3) Tobacco
(4) Cocaine

83) Tumor forming virus is known as :-

(1) Oncogenes
(2) Oncovirus
(3) Oncology
(4) Carcinogen

84) Choose correct option regarding antibodies ?

(1) IgA - helps in allergic reaction
(2) IgG - crosses placenta
(3) IgE - found in secretions
(4) IgM - exists as dimer

85) Smoking increases 'X' gase content in flood which reduces concentration of haemoglobin bound oxygen. This 'X' is

(1) CO
(2) CO_2
(3) N_2
(4) SO_2

86) The causetive agent of hepatitis B is :-

(1) RNA virus
(2) DNA virus
(3) Retrovirus
(4) Mycoplasma virus

87) BCG vaccine is used for the prevention of :-

(1) T.B
(2) Diphtheria
(3) Tetanus

(4) Typhoid

88) **Assertion (A) :-** LSD and marijuana are clinically used as analgesics.
Reason (R) :- Both these drugs suppress brain function.

(1) (A) and (R) both are correct and (R) is the correct explanation of (A)

(2) (A) and (R) both are correct but (R) is not the correct explanation of (A)

(3) Assertion is correct but reason is incorrect.

(4) Assertion and reason both are incorrect.

89) **Assertion (A) :-** Heroin obtained by acetylation of morphine.
Reason (R) :- Heroin is extracted from the latex of poppy plant papaver somniferum.

(1) (A) and (R) both are correct and (R) is the correct explanation of (A)

(2) (A) and (R) both are correct and (R) is the incorrect explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

90) Read the following statements and identify the causetive agent of these symptoms:-
(1) Lower limbs and genitals affected.
(2) Vector born
(3) Chronic inflammation of lower lymphatics.

(1) Wuchereria malayi

(2) Trichophyton

(3) Rhino virus

(4) Salmonella typhi